

# Unequal Job Opportunities and Well-Being Inequality

A Latent-Jobs Structural Decomposition

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Observed

$C_i$

Gini of consumption

single adults 0.2633    couples 0.2268

Well-being

$M_i$

Gini of the attained-bundle money metric

single adults 0.2498    couples 0.1974

low earnings    $\Leftarrow$    tastes   **or**   opportunities   **or**   resources

equivalised, household-weighted, within each population

## Compensation

unequal job opportunities should not become  
unequal well-being

## Responsibility

preference-based choices should not be  
compensated

Compensation      vs.      Responsibility

$\nexists W : \text{Full Compensation} \wedge \text{Full Responsibility}$

[Fleurbaey & Maniquet]

How much inequality in money-metric well-being is associated with unequal job opportunities rather than heterogeneous preferences, once labour supply is modelled as choice among latent jobs?

1. Do observed labour-supply choices reflect preferences alone, or also heterogeneous latent job opportunities?
2. How do conclusions differ when welfare evaluates the attained bundle versus the ex-ante opportunity prospect?
3. In a restricted P/A/B decomposition, how much welfare inequality is associated with preferences, local access, and earning opportunities?

# Inequality of opportunity

Outcome inequality



circumstances

versus

responsibility-sensitive factors

**Typical circumstances**

**Typical outcomes**

parental background · sex · ethnicity or nationality ·  
birthplace or region

earnings · income · consumption

*What share of observed inequality is associated with circumstances?*

**Italy**  
**19.5%**

Checchi & Peragine 2010

**Brazil**  
**10–37%**

Bourguignon, Ferreira & Menéndez 2007

**France**  
**12.9%**

Brunori, Ferreira & Peragine 2013

Different outcomes, circumstance sets and methods.

# From circumstances to job opportunities

|                           | Empirical EOp                              | This paper                                                              |
|---------------------------|--------------------------------------------|-------------------------------------------------------------------------|
| <b>Opportunity</b>        | observed circumstances or types            | estimated latent-job distribution                                       |
| <b>Outcome</b>            | earnings, income, consumption              | money-metric well-being                                                 |
| <b>Structural objects</b> | —                                          | preferences · access · earning opportunities                            |
| <b>Strategy</b>           | compare circumstance-defined distributions | equalise structural components, recompute welfare, decompose inequality |

**An inequality-of-opportunity question with a structural opportunity object.**

## Latent jobs

Dagsvik & Jia  
Aaberge–Colombino tradition  
Capéau, Decoster & Dekkers

## Welfare

Fleurbaey & Maniquet  
Decoster & Haan  
Jacquet, Jia & Thoresen

## Decomposition

Shorrocks · Shapley

The contribution is the structural equalisation of estimated preferences, access and earning opportunities followed by recomputation of money-metric welfare inequality.

$j = (e, k, h, w)$

4 occupation groups

$C_i(j)$  priced through EUROMOD

$$v_i(j) = L_i(j) + \beta_c \log\left(\frac{C_i(j)}{\lambda_c}\right)$$

$L_i$

value of leisure: age; children for  
women

$\beta_c$

weight on log consumption,  
estimated

$\lambda_c$

units convention

## Opportunities: the choice probability

$$P_i(j) = \frac{\exp\{v_i(j)\} g_i(j)}{\int \exp\{v_i(k)\} g_i(k) d\nu(k)}$$

**access**

employment mass

**hours**

bands

**occupation**

by sex

**wage offers**

education, experience

valued   or   plentiful

# How are preferences and opportunities separated?

|                            |                                                                 |
|----------------------------|-----------------------------------------------------------------|
| <b>Preferences</b>         | smooth systematic utility over hours                            |
| <b>Access</b>              | local unemployment and geography, excluded from utility         |
| <b>Hours opportunities</b> | band structure                                                  |
| <b>Wage opportunities</b>  | education and experience shift the offer distributions          |
| <b>Budgets</b>             | EUROMOD generates heterogeneous nonlinear consumption schedules |

**These restrictions give different empirical variation to preferences and opportunities.**

## EU-SILC France

collected 2016, incomes 2015

1,540 single-adult households

2,223 couple households

## EUROMOD

French 2015 policy system

$h \in [5, 70]$      $w \in [2, 590]$

100 priced alternatives per household

## Household-specific opportunities

free coordinates 41

population fit 0.0140

## Common opportunities

free coordinates 10 / 16

population fit 0.0273 / 0.0264

criterion worse by at least 141.64

choices reflect opportunities, not preferences alone

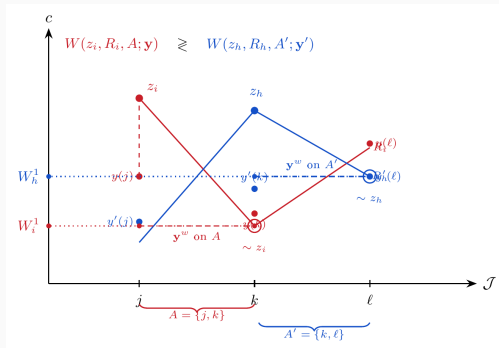
single adults; population fit is the mean absolute deviation of predicted from observed moments

## Selected estimates

|                                                      | Single adults    | Couples          |
|------------------------------------------------------|------------------|------------------|
| $\beta_c$ consumption weight                         | 2.0387 (0.2917)  | 2.1017 (0.2939)  |
| $\beta_{E,s}$ local unemployment exposure, in access | -1.4422 (0.2356) | -1.1924 (0.1557) |
| $\beta_{wH}$ high education, wage-offer location     | 0.1491 (0.0308)  | 0.1817 (0.0197)  |
| $\sigma$ log-wage dispersion                         | 0.3815 (0.0133)  | 0.3631 (0.0072)  |

cluster-robust standard errors in parentheses; the two populations are estimated separately

# Welfare: attained outcomes versus job prospects



Own-set equal-consumption equivalents: each individual gets a common consumption on every job in their own ability set; the level at which the preferred job is indifferent to the attained bundle is their money metric, comparable across individuals. Adapted from Haydar and Maniquet (2026), work in progress.

*What constant consumption over the household's own job environment is equivalent to its prospect?*

$$J_i = \int e^{L_i(j)} \left( \frac{C_i(j)}{\lambda_c} \right)^{\beta_c} \widehat{g}_i(j) d\nu(j), \quad H_i = \int e^{L_i(j)} \widehat{g}_i(j) d\nu(j),$$

$$M_i^{\text{EA}} = \lambda_c \exp \left\{ \frac{\log J_i - \log H_i}{\beta_c} \right\}$$

$J_i$

the actual prospect

$H_i$

same jobs, equal pay

$M_i^{\text{EA}}$

flat-consumption equivalent

*What is the money equivalent of the attained bundle?*

$$M_i^{\text{att}} = C_i^{\text{obs}} \exp \left\{ \frac{L_i(j_i^{\text{obs}}) - L_i(o)}{\beta_c} \right\}$$

$o$

non-employment, reachable by all

$j_i^{\text{obs}}$

the only route for opportunities

# Inequality and Shapley decomposition

**P**

preferences: age, children

**A**

local unemployment  
exposure, region, urban  
or rural location, year

**B**

wage-offer location

**held fixed**

household resources,  
needs and composition

$$\phi_k^P = \sum_{S \subseteq \{P, A, B\} \setminus \{k\}} \frac{|S|! (3 - |S| - 1)!}{3!} \left[ v^P(S \cup \{k\}) - v^P(S) \right]$$

$$v^P(S) = I^P(\emptyset) - I^P(S) \quad I^P: \text{household-weighted Gini of } M^P$$

# How much is associated with opportunities?

## Ex-ante prospect

equivalised

Single adults

20.3%

Couples

7.9%

## Attained bundle

equivalised

raw 14.8%

1.9%

raw 21.3%

3.5%

raw 2.4%

raw 6.7%

**A + B as % of the relevant baseline Gini**

The two measures also differ in aggregation: the prospect measure inverts within each household; the attained-bundle measure averages inequality across simulated realisations.

# Which opportunity channel matters?

## Single adults

access  
15.5%

earning opportunities

4.8%

$A + B = 20.3\%$

## Couples

access  
3.7%

earning opportunities

4.2%

$A + B = 7.9\%$

prospective welfare, equivalised: % of baseline Gini

single adults: access > earnings      couples: earnings > access

attained-bundle welfare: earnings > access for both populations

# Where does baseline inequality go?

shares of baseline Gini · prospect welfare, equivalised

|                                                                                     | Single adults | Couples |
|-------------------------------------------------------------------------------------|---------------|---------|
| Preferences $\phi_P/l_0$                                                            | 0.1%          | -1.8%   |
| Local access $\phi_A/l_0$                                                           | 15.5%         | 3.7%    |
| Earning opportunities $\phi_B/l_0$                                                  | 4.8%          | 4.2%    |
| Other / outside current P-A-B decomposition<br>$1 - (\phi_P + \phi_A + \phi_B)/l_0$ | 79.6%         | 93.9%   |
| Total                                                                               | 100.0%        | 100.0%  |

Couples: A + B alone = 7.9%; P + A + B = 6.1% because the preference contribution is negative (-1.8%).

Current decomposition equalises P, A and B only.

# What is not claimed

## Scope

Household resources, needs and composition  
held fixed

Access: local access, not total opportunity

Preliminary decomposition

## Not claimed

Not causal

No parameter uncertainty yet

Preferences are not responsibility

Neither perspective is designated primary

prospects versus attained outcomes: two different welfare questions

1. Choices reflect opportunities, not preferences alone.
2. Prospects: access leads for single adults. Attained outcomes: earnings lead. Couples: earnings lead under both.
3. Opportunities account for a bounded part of welfare inequality; the preference contribution changes sign with equalisation.

Next: extend the decomposition to household resources and needs.

**Thank you**

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# Appendix map

|               |                               |                          |                         |                    |
|---------------|-------------------------------|--------------------------|-------------------------|--------------------|
| Literature    | EOp benchmarks                | Literature               |                         |                    |
| Model         | Utility specification         | Opportunity density      | Likelihood              | Identification     |
|               | Observational equivalence     |                          |                         |                    |
| Estimation    | Preferences                   | Employment access        | Hours and occupation    | Wage offers        |
|               | Common opportunities          | Inference                | Parameter uncertainty   |                    |
| Fit           | Extensive margin              | Fit by margin            | Thirty-seven-hour point | Matched households |
|               | Marginal rate of substitution |                          |                         |                    |
| Welfare       | ATT derivation                | EA derivation            | EA validation           | Equivalisation     |
|               | Prospect reference            | ATT numerical design     | Aggregation orders      | Theory scope       |
| Decomposition | Central figure                | EA Gini points           | ATT Gini points         | EA channel shares  |
|               | ATT channel shares            | Composition within P/A/B | Shapley vs one-factor   | Outside P-A-B      |
|               | Robustness                    |                          |                         |                    |
| Data          | Sample selection              | EUROMOD pricing          | Numerical design        |                    |

## Backup: inequality-of-opportunity benchmarks

|               | share  | outcome              | data           | measure                   | source                              |
|---------------|--------|----------------------|----------------|---------------------------|-------------------------------------|
| Italy         | 19.5%  | gross earnings       | SHIW 1993–2000 | MLD; parental education   | Checchi & Peragine (2010), p. 445   |
| Brazil, urban | 10–37% | male hourly earnings | PNAD 1996      | Theil; five circumstances | Bourguignon et al. (2007), abstract |
| France        | 12.9%  | post-tax earnings    | EU-SILC 2005   | MLD; five circumstances   | Brunori et al. (2013), Table 1      |

### These benchmarks

shares of earnings inequality attributed to observed circumstances

### This paper

a share of welfare inequality from estimated latent opportunity components, resources and needs held fixed

## Choice sets and restrictions

Aaberge, Colombino & Wennemo (2009): alternative representations of the choice set

Beffy, Blundell, Bozio, Laroque & Tô (2019): labour supply with restricted choices

## Latent jobs and welfare

Aaberge, Dagsvik & Strøm;  
Dagsvik & Strøm; Dagsvik & Jia;  
Capéau, Decoster & Dekkers  
Fleurbay & Maniquet; Decoster  
& Haan; Bargain et al.; Jacquet,  
Jia & Thoresen

## Inequality of opportunity and decomposition

Roemer; Checchi & Peragine;  
Bourguignon, Ferreira &  
Menéndez; Brunori, Ferreira &  
Peragine  
Shorrocks; Sastre & Trannoy;  
Creedy & Hérault; Mühlhan

## Backup: observational equivalence

### Without restrictions

a taste against certain hours and a scarcity of jobs at those hours predict the same choices

### With restrictions

smooth tastes, banded availability and excluded access shifters move different parts of the choice distribution

If availability heterogeneity is omitted, some of its effects can be absorbed by the estimated utility component.

maintained restrictions, not tested · regional variation is not a causal design

# Backup: attained-bundle numerical design

## Sample

the already-priced estimation  
panel; no re-estimation and no  
new pricing

## Counterfactuals

equalised covariates set to  
household-weighted means;  
attained bundle simulated by  
Gumbel-max with common  
random numbers

## Checks

replications at fixed estimates; an  
independent second seed; exact  
Shapley closure

short weekly hours are sparsely represented in the integration panel

Monte Carlo ranges are simulation spreads, not confidence intervals

## Backup: two aggregation orders

### Attained bundle

Gini within each simulated realisation, then averaged

$$I^{\text{att}}(S) = \frac{1}{R} \sum_r \mathcal{G}_w \left( \{M_{i,r}^{\text{att}}(T_S x_i)\}_i \right)$$

$r$ : simulated realisation of attained bundles, common random numbers across coalitions

### Prospect

inverted within each household, then one Gini

$$I^{\text{EA}}(S) = \mathcal{G}_w \left( \{M_i^{\text{EA}}(T_S x_i)\}_i \right)$$

$M_i^{\text{EA}} = \lambda_c \exp\{(\log J_i - \log H_i)/\beta_c\}$ , integrated over the household's prospect

the average of realised inequality versus the inequality of household equivalents

## Backup: the prospect reference

$$M_i^{\text{EA}} = \left( \int C_i(j)^{\beta_c} \omega_i(j) d\nu(j) \right)^{1/\beta_c}, \quad \omega_i(j) = \frac{e^{L_i(j)} \hat{g}_i(j)}{H_i}$$

### Own-opportunity reference

the reference prospect keeps the household's own jobs, availability and preferences, and equalises pay

### Weighted generalised mean

the metric is a generalised mean of consumption with exponent  $\beta_c$ , weighted by leisure value and availability

the reference moves with the household's opportunity environment

### What the companion theory does and does not cover

#### Covered

The deterministic own-set equal-consumption construction is characterised axiomatically in the companion paper.

#### Not covered

The stochastic extension to a prospect is this paper's own step and is not characterised there.

### The asymmetry

The attained-bundle measure uses systematic utility only, while the prospect measure's expected-maximum construction makes the taste component relevant to the prospect's value.

Whether that asymmetry is the right normative choice is open.

## Backup: parameter uncertainty

### Estimated

cluster-robust standard errors for  
every estimated coordinate,  
clustered on the household

### Not yet propagated

estimation uncertainty has not  
been carried through either welfare  
decomposition

### Simulation checks

second seeds and repeat runs  
measure numerical noise, not  
sampling uncertainty

the decomposition shares are point estimates without confidence intervals

## Backup: utility specification

$$v_i(j) = L_i(j) + \beta_c \log\left(\frac{C_i(j)}{\lambda_c}\right), \quad L_i(j) = \beta_\ell^g(\mathbf{x}_i) \mathcal{B}(\tilde{\ell}_i(j); \theta_\ell^g)$$

$$\tilde{\ell}_i(j) = \frac{80 - h}{10}, \quad \mathcal{B}(z; \theta) = \frac{z^\theta - 1}{\theta}$$

$\beta_\ell^g(\mathbf{x}_i)$

intercept, centred age and its  
square (units of 10 years)

$k_i$

children under 20, women only

couples

sum of the two spouses' leisure  
terms

## Backup: opportunity density

$$g_i(o) = 1, \quad g_i(j) = g_i^E g_i^H(h) g_i^{\text{Occ}}(k) g_i^W(w | k) \quad \text{for market jobs } j \neq o$$

### access

$g_i^E$ : local unemployment exposure, region, urban or rural location, year

### hours

$g_i^H$ : elevated bands over a residual set

### occupation

$g_i^{\text{Occ}}$ : sex-specific group masses

### wage offers

$g_i^W$ : log-normal, truncated to [2, 590]

wage-offer location  $\mu_i(k)$ : education, potential experience and its square, occupation

## Backup: sampled-alternative likelihood

$$V_{ij} = v_i(j) + \log g_i(j) - \log q_{ij}$$

$$\Pr(y \mid \mathcal{C}_i) = \frac{n_y \exp V_{iy}}{\sum_{s \in \mathcal{C}_i} \exp V_{is}}$$

$q_{ij}$

proposal density, fitted out of fold

$\mathcal{C}_i$

observed job and 100 draws: 101  
rows

$n_y$

multiplicity of a repeated draw

# Backup: identification

## Functional form

smooth preferences in hours;  
banded hours opportunities

## Exclusion

local unemployment exposure and  
geography shift access, not utility

## Independence

wage offers independent of hours,  
given occupation

maintained, not tested · not a causal design

## Backup: preference parameters

|                                 | Single men                                       | Single women                                       | Couple men                                      | Couple women                                      |
|---------------------------------|--------------------------------------------------|----------------------------------------------------|-------------------------------------------------|---------------------------------------------------|
| leisure weight, intercept       | 8.5199 (3.5480)<br><small>beta_l0_sm</small>     | 5.8683 (2.1346)<br><small>beta_l0_sf</small>       | 3.9814 (0.7466)<br><small>beta_l0_m</small>     | 13.1029 (4.7760)<br><small>beta_l0_f</small>      |
| leisure weight, age             | 1.4820 (1.0236)<br><small>beta_l_age_sm</small>  | 0.0678 (0.4792)<br><small>beta_l_age_sf</small>    | -0.0080 (0.0287)<br><small>beta_l_age_m</small> | -0.1228 (0.1260)<br><small>beta_l_age_f</small>   |
| leisure weight, age squared     | 0.7783 (0.7732)<br><small>beta_l_age2_sm</small> | 1.0000 (at bound)<br><small>beta_l_age2_sf</small> | 0.0065 (0.0030)<br><small>beta_l_age2_m</small> | 0.0073 (0.0103)<br><small>beta_l_age2_f</small>   |
| leisure weight, children        | —                                                | 0.1666 (0.4422)<br><small>beta_l_nkids_sf</small>  | —                                               | -0.3055 (1.1472)<br><small>beta_l_nkids_f</small> |
| leisure curvature $\theta_\ell$ | -1.6263 (0.3301)<br><small>theta_l_sm</small>    | -0.9274 (0.2161)<br><small>theta_l_sf</small>      | -0.9761 (0.1357)<br><small>theta_l_m</small>    | -1.6863 (0.2458)<br><small>theta_l_f</small>      |
| consumption weight $\beta_c$    | 2.0387 (0.2917)<br><small>beta_c</small>         |                                                    | 2.1017 (0.2939)<br><small>beta_c</small>        |                                                   |

cluster-robust standard errors; parameter identifiers in grey; the male children term in couples is structurally zero

## Backup: employment-access parameters

|                             | identifier                 | Singles          | Couples                                        |
|-----------------------------|----------------------------|------------------|------------------------------------------------|
| employment constant         | beta_E; beta_E_m, beta_E_f | −3.1735 (0.4053) | men −2.1233 (0.3195)<br>women −2.8148 (0.3021) |
| local unemployment exposure | beta_E_gsur                | −1.4422 (0.2356) | −1.1924 (0.1557)                               |
| densely populated           | beta_E_drgur               | −0.0288 (0.2209) | −0.1823 (0.1657)                               |
| intermediate density        | beta_E_drgmd               | 0.0641 (0.2614)  | −0.4174 (0.1861)                               |
| Region 2                    | beta_E_drgn2               | −0.3780 (0.3292) | −0.1546 (0.2442)                               |
| Region 3                    | beta_E_drgn3               | −0.1117 (0.3891) | 0.0873 (0.2808)                                |
| Region 4                    | beta_E_drgn4               | −0.8218 (0.3810) | 0.0150 (0.3068)                                |
| Region 5                    | beta_E_drgn5               | −0.5162 (0.3274) | −0.1455 (0.2555)                               |
| Region 6                    | beta_E_drgn6               | −0.7222 (0.3528) | −0.2844 (0.2748)                               |
| Region 7                    | beta_E_drgn7               | −0.5374 (0.3503) | −0.1396 (0.2651)                               |
| Region 8                    | beta_E_drgn8               | −0.4512 (0.3401) | −0.1659 (0.2660)                               |

references: the first region and thinly populated areas; couples share the shifters across spouses

## Backup: hours and occupation opportunities

| Hours bands      |            |  |              |  |              |
|------------------|------------|--|--------------|--|--------------|
|                  | Singles    |  | Couple men   |  | Couple women |
| part-time lower  | 0.0298     |  | −1.1094      |  | −0.5173      |
|                  | beta_h_pt1 |  | beta_h_pt1_m |  | beta_h_pt1_f |
| part-time upper  | 0.7080     |  | 0.0835       |  | 1.1586       |
|                  | beta_h_pt2 |  | beta_h_pt2_m |  | beta_h_pt2_f |
| narrow full-time | 2.0656     |  | 2.2888       |  | 2.0284       |
|                  | beta_h_f35 |  | beta_h_f35_m |  | beta_h_f35_f |
| full-time upper  | 1.9387     |  | 2.3801       |  | 1.6825       |
|                  | beta_h_ft  |  | beta_h_ft_m  |  | beta_h_ft_f  |
| long hours       | −0.0834    |  | 0.6977       |  | −0.3109      |
|                  | beta_h_lh  |  | beta_h_lh_m  |  | beta_h_lh_f  |

| Occupation masses |              |              |              |              |
|-------------------|--------------|--------------|--------------|--------------|
|                   | Singles      |              | Couples      |              |
|                   | men          | women        | men          | women        |
| Occupation 2      | −1.3737      | 0.1135       | −1.5097      | 0.2074       |
|                   | beta_occ_2_m | beta_occ_2_f | beta_occ_2_m | beta_occ_2_f |
| Occupation 3      | −2.1096      | −0.4060      | −2.2532      | −0.1804      |
|                   | beta_occ_3_m | beta_occ_3_f | beta_occ_3_m | beta_occ_3_f |
| Occupation 4      | −0.3259      | 0.5632       | 0.1594       | 0.8159       |
|                   | beta_occ_4_m | beta_occ_4_f | beta_occ_4_m | beta_occ_4_f |

estimates with identifiers; standard errors in the paper

## Backup: wage-offer equation

|                              | identifier   | Singles          | Couples          |
|------------------------------|--------------|------------------|------------------|
| intercept                    | beta_w0      | 2.0138 (0.0561)  | 2.0480 (0.0335)  |
| low education                | beta_w_educL | 0.0566 (0.0372)  | −0.0433 (0.0217) |
| high education               | beta_w_educH | 0.1491 (0.0308)  | 0.1817 (0.0197)  |
| potential experience         | beta_w_pexp  | 0.2408 (0.0862)  | 0.5643 (0.0546)  |
| experience squared           | beta_w_pexp2 | −0.0282 (0.0390) | −0.1598 (0.0243) |
| Occupation 2                 | delta_occ_2  | −0.0347 (0.0399) | −0.0740 (0.0231) |
| Occupation 3                 | delta_occ_3  | 0.0622 (0.0383)  | 0.0394 (0.0228)  |
| Occupation 4                 | delta_occ_4  | 0.2785 (0.0368)  | 0.2146 (0.0223)  |
| log-wage dispersion $\sigma$ | sigma        | 0.3815 (0.0133)  | 0.3631 (0.0072)  |

medium education is the reference; experience in units of 20 years

## Backup: common versus household-specific opportunities

| Single adults                                 | free coordinates | criterion | population fit |
|-----------------------------------------------|------------------|-----------|----------------|
| household-specific opportunities              | 41               | 6253.463  | 0.0140         |
| common opportunity distribution               | 10               | 6403.974  | 0.0273         |
| common shape; employment and hours in utility | 16               | 6395.108  | 0.0264         |

same households, sampled alternatives and criterion; not converted into a formal test

# Backup: curvature and inference

## Single adults

free coordinates 41, interior 40

criterion 6253.463

smallest Hessian eigenvalue 0.1669

## Couples

free coordinates 47, interior 47

criterion 10283.034

smallest Hessian eigenvalue 0.0771

cluster-robust standard errors on the household · eigenvalues on the interior block

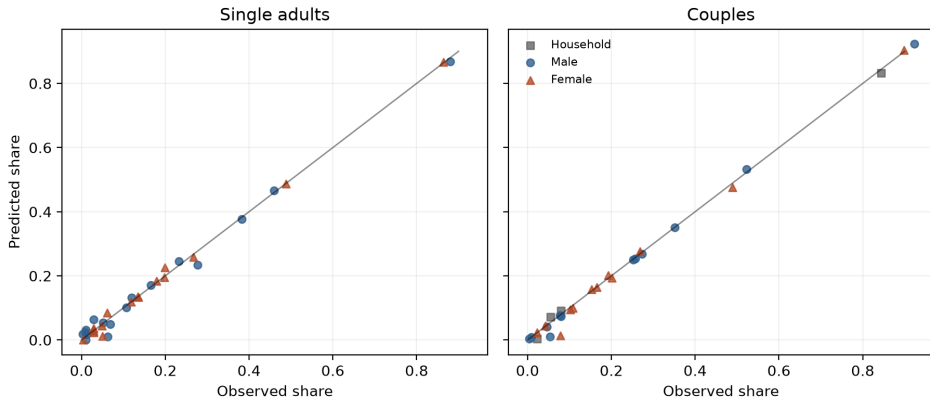
## Backup: extensive-margin fit

|               | weighted accuracy | model-simulated range | integration ratio |
|---------------|-------------------|-----------------------|-------------------|
| single women  | 85.7%             | 80.0–85.9%            | 0.096             |
| coupled women | 89.8%             | 88.1–91.2%            | 0.049             |
| single men    | withheld          | —                     | 0.361             |
| coupled men   | withheld          | —                     | 0.353             |

withheld where numerical integration error is too large relative to sampling variation

# Backup: fit by margin

Corrected population moments: observed versus predicted



## Backup: the thirty-seven-hour point

single men

5.2

pp

single women

3.8

pp

coupled men

4.4

pp

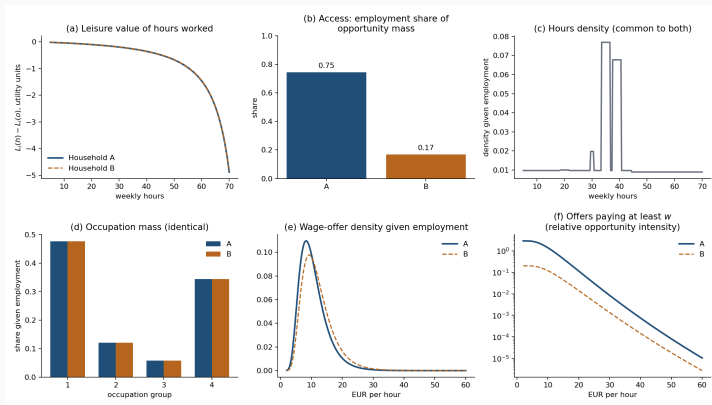
coupled women

6.5

pp

observed minus predicted share at the observed thirty-seven-hour mass point

# Backup: matched households

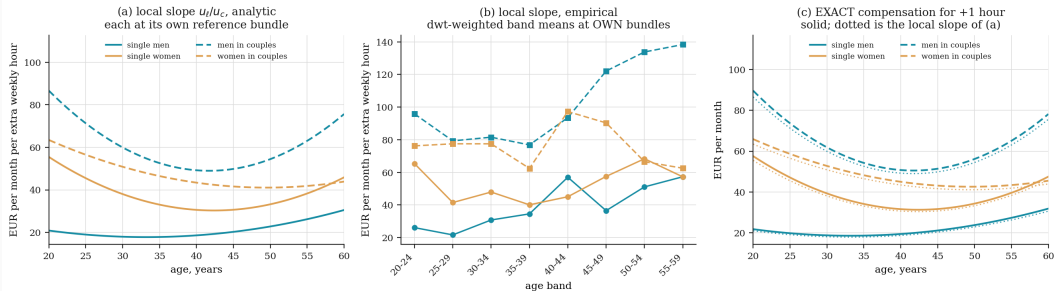


access:  $A = 14.5 \times B$

wage-offer location:  $B + 11.5$  log points

# Backup: marginal rate of substitution

The local compensation  $u_l/u_c$  by age and sex, and the exact compensation for one more weekly hour



## Backup: attained-bundle derivation

$$L_i(o) + \beta_c \log\left(\frac{M_i^{\text{att}}}{\lambda_c}\right) = L_i(j_i^{\text{obs}}) + \beta_c \log\left(\frac{C_i^{\text{obs}}}{\lambda_c}\right)$$

$$\implies M_i^{\text{att}} = C_i^{\text{obs}} \exp\left\{\frac{L_i(j_i^{\text{obs}}) - L_i(o)}{\beta_c}\right\}$$

### Reference job

the preferred job when every job pays the same:  
here non-employment for every household

### Staying-home equivalent

a verified property of this specification, not a  
general theorem

## Backup: ex-ante derivation

$$J_i = \int_{\Omega_i} e^{L_i(j)} \left( \frac{C_i(j)}{\lambda_c} \right)^{\beta_c} \widehat{g}_i(j) d\nu(j)$$

$$J_i^{\text{ref}}(m) = \left( \frac{m}{\lambda_c} \right)^{\beta_c} H_i, \quad H_i = \int_{\Omega_i} e^{L_i(j)} \widehat{g}_i(j) d\nu(j)$$

$$J_i^{\text{ref}}(M_i^{\text{EA}}) = J_i \quad \implies \quad M_i^{\text{EA}} = \lambda_c \exp \left\{ \frac{\log J_i - \log H_i}{\beta_c} \right\}$$

a common rescaling of  $\widehat{g}_i$  cancels  $\cdot \log J_i$ : expected utility of the best reachable job, up to a constant

# Backup: ex-ante numerical validation

## Identities

constant-consumption identity

one-state identity

common intensity-rescaling invariance

flat-reference inversion

exact Shapley adding-up

## Numerics

exact reference integral on the priced set

effective sample size and weight concentration

two-seed stability of welfare, Gini and Shapley

finite positive welfare for every household

independent reimplementations

all passed for both populations · a correct computation, not causal identification or parameter uncertainty

## Backup: equivalisation

$$M_i^{p,eq} = \frac{M_i^p}{e_i}, \quad e_i : \text{modified-OECD scale}$$

**Couples, ex ante: A + B**

21.3%  $\longrightarrow$  7.9%

raw  $\rightarrow$  equivalised; % of baseline Gini

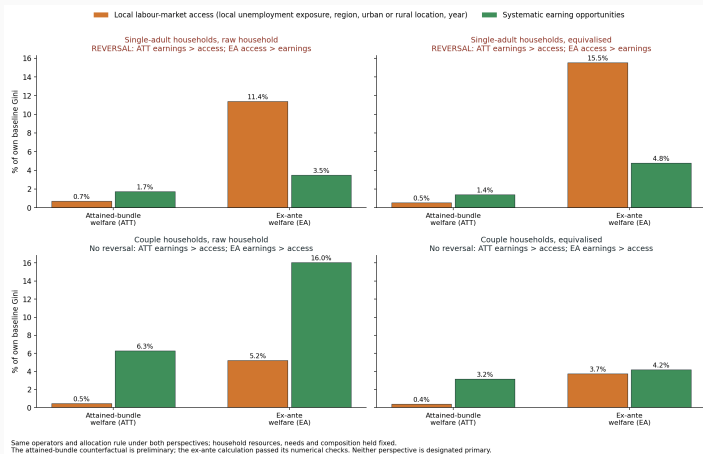
**Couples, ex ante: preferences**

0.00291  $\longrightarrow$  -0.00212

Gini points, raw  $\rightarrow$  equivalised

a normative convention, reported both ways; populations never combined or compared

# Backup: central figure



## Backup: ex-ante decomposition in Gini points

|               | scale       | baseline Gini | P        | A       | B       | A + B, % of baseline | ordering          |
|---------------|-------------|---------------|----------|---------|---------|----------------------|-------------------|
| Single adults | raw         | 0.1595        | 0.00157  | 0.01812 | 0.00555 | 14.8                 | access > earnings |
|               | equivalised | 0.1406        | 0.00020  | 0.02183 | 0.00673 | 20.3                 | access > earnings |
| Couples       | raw         | 0.1277        | 0.00291  | 0.00667 | 0.02049 | 21.3                 | earnings > access |
|               | equivalised | 0.1200        | -0.00212 | 0.00449 | 0.00504 | 7.9                  | earnings > access |

exact Shapley contributions in Gini points; household-weighted

## Backup: attained-bundle decomposition in Gini points

|               | scale       | baseline Gini | P       | A      | B      | A + B, % of baseline | ordering          |
|---------------|-------------|---------------|---------|--------|--------|----------------------|-------------------|
| Single adults | raw         | 0.2337        | 0.0012  | 0.0016 | 0.0040 | 2.4                  | earnings > access |
|               | equivalised | 0.2451        | -0.0003 | 0.0013 | 0.0034 | 1.9                  | earnings > access |
| Couples       | raw         | 0.2035        | 0.0064  | 0.0009 | 0.0128 | 6.7                  | earnings > access |
|               | equivalised | 0.1971        | -0.0019 | 0.0007 | 0.0062 | 3.5                  | earnings > access |

exact Shapley contributions in Gini points; preliminary; household-weighted

## Backup: ex-ante channels, % of baseline

|               |             | P      |          |            |          | A          |          | B          |  |
|---------------|-------------|--------|----------|------------|----------|------------|----------|------------|--|
|               | scale       | $I_0$  | Gini pts | % of $I_0$ | Gini pts | % of $I_0$ | Gini pts | % of $I_0$ |  |
| Single adults | raw         | 0.1595 | 0.00157  | 0.99       | 0.01812  | 11.37      | 0.00555  | 3.48       |  |
|               | equivalised | 0.1406 | 0.00020  | 0.14       | 0.02183  | 15.52      | 0.00673  | 4.79       |  |
| Couples       | raw         | 0.1277 | 0.00291  | 2.28       | 0.00667  | 5.22       | 0.02049  | 16.04      |  |
|               | equivalised | 0.1200 | −0.00212 | −1.76      | 0.00449  | 3.75       | 0.00504  | 4.20       |  |

each contribution divided by its own baseline Gini  $I_0$ ; signed; the rows are not forced to sum to one hundred, because the remainder is outside the current P-A-B decomposition

## Backup: attained-bundle channels, % of baseline

|               |             | P      |          |            | A        |            | B        |            |
|---------------|-------------|--------|----------|------------|----------|------------|----------|------------|
|               | scale       | $I_0$  | Gini pts | % of $I_0$ | Gini pts | % of $I_0$ | Gini pts | % of $I_0$ |
| Single adults | raw         | 0.2337 | 0.0012   | 0.53       | 0.0016   | 0.69       | 0.0040   | 1.73       |
|               | equivalised | 0.2451 | -0.0003  | -0.13      | 0.0013   | 0.54       | 0.0034   | 1.40       |
| Couples       | raw         | 0.2035 | 0.0064   | 3.16       | 0.0009   | 0.45       | 0.0128   | 6.28       |
|               | equivalised | 0.1971 | -0.0019  | -0.98      | 0.0007   | 0.37       | 0.0062   | 3.15       |

each contribution divided by its own baseline Gini  $I_0$ ; signed; preliminary

## Backup: composition of the explained component

$$s_k = \frac{\phi_k}{\phi_P + \phi_A + \phi_B}, \quad k \in \{P, A, B\}$$

| ex-ante prospect, equivalised | Single adults | Couples |
|-------------------------------|---------------|---------|
| Preferences $s_P$             | 0.7%          | −28.5%  |
| Local access $s_A$            | 75.9%         | 60.6%   |
| Earning opportunities $s_B$   | 23.4%         | 67.9%   |
| Total                         | 100.0%        | 100.0%  |

These are shares of the currently explained P/A/B change, not shares of total inequality. Negative components are possible under Shapley attribution.

## Backup: Shapley versus one-factor equalisation

$$\text{one-factor: } v^P(\{k\}) = I^P(\emptyset) - I^P(\{k\})$$

$$\text{Shapley: } \phi_k^P = \text{average over all orders of } v^P(S \cup \{k\}) - v^P(S)$$

$$\phi_P^P + \phi_A^P + \phi_B^P = I^P(\emptyset) - I^P(\{P, A, B\})$$

one-factor effects need not add up   ·   Shapley contributions do, exactly

## Backup: outside the current P-A-B decomposition

### Held fixed

household resources, needs and  
composition

### Retained

sex-specific and  
household-type-specific preference  
and opportunity blocks

### Not equalised

wage-draw luck and the common  
offer spread

wage-offer location differences: at most about 0.10 log points

common offer spread: about 0.37 log points

next: extend the decomposition to household resources and needs, after fixing the appropriate operator

# Backup: robustness of the attained-bundle decomposition

## Second simulation seed

every coalition Gini and contribution  
reproduces closely, including the preference  
sign change

## Observed job in the simulation

attained under every coalition by 2.9% of single  
adults and 3.9% of couples  
excluding it moves the total change by at most  
9.3%, with no sign change

simulation checks, not confidence intervals

## Backup: sample selection

| screen                                                                                  | single-adult | couple       |
|-----------------------------------------------------------------------------------------|--------------|--------------|
| the raw France 2016 input file                                                          | 4,038        | 5,965        |
| one- or two-adult households (composition screen)                                       | 4,038        | 5,965        |
| every adult aged 20 to 60                                                               | 2,131        | 3,662        |
| no adult still in education                                                             | 2,036        | 3,521        |
| no old-age, disability or survivor pension receipt                                      | 1,755        | 3,218        |
| labour status in scope                                                                  | 1,598        | 2,412        |
| no other employable or earning member                                                   | 1,564        | 2,323        |
| hours and wage inside the modelled support                                              | 1,555        | 2,275        |
| hours outside [5,70] or unsupported military occupation (ISCO 0) on an employed decider | 1,543        | 2,223        |
| observed chosen alternative priced at non-positive disposable consumption               | 1,540        | 2,223        |
| <b>estimation sample</b>                                                                | <b>1,540</b> | <b>2,223</b> |

## Backup: EUROMOD pricing

### Inputs

a work arrangement: hours, hourly wage, earnings split; household roster and non-labour inputs

### EUROMOD

French 2015 policy system; only deciders receive counterfactual overrides

### Output

$C_i(j)$ : household disposable income, summed over all members

every sampled alternative and every welfare integration point is priced  
tax-benefit accounting identity holds to machine precision

## Estimation

101

rows per household: the observed  
job and 100 sampled alternatives

## Predictive fit

2,048

common integration points per  
household

## Supports

[5, 70]

weekly hours; wages in [2, 590] per  
hour

sampling devices are computational: they are not offers and enter no welfare measure